1. The anterior surface of the midbrain is formed by the [appeared on 2002/02]
a) Tectum
b) Tegmentum
c) Superior cerebellar peduncle
d) Reticular formation
e) Cerebellar peduncle

2. Truncal ataxia is seen in patients with lesions of the [appeared on 2002/02]
a) VPL of the thalamus
b) Medial longitudinal fasciculus (MLF)
c) Vermis
d) Pons
e) Spinal cord

3. It forms the roof of the 3rd ventricle
a) Corpus callosum
b) Lamina terminalis
c) Cerebellar aqueduct
d) Tuber cinereum
e) Fornix

4. It is the most frequent location of hypertensive hemorrhages
a) Cerebellar cortex
b) Globose nucleus
c) Dentate nucleus
d) Floor of the 4th ventricle
e) Vermis

5. Which cranial nerve is the most commonly affected in patients with a cranial base fracture affecting the anterior cranial fossa
a) Optic nerve
b) Ophthalamic nerve
c) Olfactory nerve
d) Oculomotor nerve
e) Abducens nerve

6. It is located between the globus pallidus and the external capsule [appeared on 2002/02]
a) Internal capsule
b) Subthalamus
c) Substantia nigra
d) Claustrum
e) Putamen

7. After entering the spinal cord the fibers that convey pain and temperature run in
a) Clarke's tract
b) Fasciculus cuneatus
c) Fasciculus gracilis
d) Lissauer's tract
e) Flesching's tract

8. It becomes continuous with the amygdala
a) Putamen
b) Internal capsule
c) Caudate
d) External capsule
e) Globus pallidus

9. Rigidity is commonly seen in patients with [appeared on 2002/02]
a) Spinal cord damage
b) Cerebellar damage
c) Brainstem damage
d) Increased intracranial pressure
e) Basal ganglia damage

10. Muscle contraction in response to a stimulus involves activation of the alpha, beta and gamma motorneurons of lamina
a) VI
b) X
c) VII
d) IV
e) IX

24. The loop of Meyer is part of the
a) Optic nerve
b) Optic chiasm
c) Optic tract
d) Optic radiation
e) Visual cortex

25. The most frequent clinical picture of patients with pineal region tumors is
a) Medullary medical sector syndrome
b) Foville's syndrome
c) Weber's syndrome
d) Parinaud's syndrome
e) Benedikt's syndrome

26. It's fibers form the afferent limb of the corneal reflex
a) Ophthalmic nerve
b) Facial nerve
c) Maxillary nerve
d) Mandibular nerve
e) Glossopharyngeal nerve

27. Lesions of the medial longitudinal fasciculus results in [know the medial longitudinal fasciculus]
a) Internuclear ophthamoplegia
b) Weber's syndrome
c) Parinaud's syndrome
d) Dejerine-Roussy syndrome
e) Wallenberg's syndrome

28. Occlusion of the posterioinferior cerebellar artery results in
a) Foville's syndrome
b) Benedikt's syndrome
c) Wallenberg's syndrome
d) Parinaud's syndrome
e) Weber's syndrome

29. It is mainly excitatory to the extensor alpha motoneurons [no answer yet]
a) Corticobulbar tract
b) Corticotectal tract
c) Spinothalamic tract
d) Fasciculus gracilis
e) Reticulospinal tract

30. Its SVE fibers innervate the stylohyoid muscle
a) Facial
b) Trigeminal
c) Glossopharyngeal
d) Vagus
e) Accessory

31. After decussating, fibers from all three acoustic striae join the [appeared on 2002/02]
a) Medial lemniscus
b) Trigeminal lemniscus
c) Lateral lemniscus
d) Posterior lemniscus
e) Anterior lemniscus

32. It is part of the efferent limb of the gag reflex [appeared on 2002/02]
a) Intermediate nerve
b) Glossopharyngeal nerve
c) Trigeminal nerve
d) Vagus nerve
e) Accessory nerve

33. A giant anuerysm of the superior cerebellar artery can compress the
a) Abducens
b) Trochlear
c) Oculomotor
d) Facial
e) Glossopharyngeal

36. Features of Horner's syndrome include [appeared on 2002/02]
a) Ptosis, miosis, hyperhidrosis
b) Ptosis, miosis, absence of corneal reflex
c) Ptosis, miosis, anhydrosis
d) Ptosis, dilation of the pupil, anhydrosis
e) Ptosis, dilation of the pupil, hyperhydrosis

37. Which of the following ganglia gives rise to the carotid plexus
a) Superior cervical
b) Middle cervical
c) Stellate
d) Inferior cervical
e) First thoracic

38. A cortical lesion affecting the striate cortex produces [appeared on 2002/02]
a) Ipsilateral blindness
b) Contralateral blindness
c) Heteronymous hemianopsia
d) Homonymous hemianopsia
e) Loss of pupillary reflexes

39. The stellate ganglion is formed by the [appeared on 2002/02]
a) Superior and middle cervical ganglia
b) Superior and inferior cervical ganglia
c) Middle and inferior cervical ganglia
d) Inferior cervical and first thoracic ganglia
e) Middle cervical and first thoracic ganglia

40. It is the terminal branch of the basilar artery
a) Anterior cerebral
b) Middle cerebral
c) Posterior cerebral
d) Anterior cerebellar
e) Posteroinferior cerebellar

41. Inferior sagittal sinus
a) It receives blood from the middle superficial cerebral vein
b) Lies in the attached border of the cerebellar falx
c) Drains into the transverse sinus
d) Runs in the free edge of the cerebral falx
e) Drains into the internal jugular vein

42. Cavernous sinus
a) It receives blood from the middle superficial cerebral vein
b) Lies in the attached border of the cerebellar falx
c) Drains into the transverse sinus
d) Runs in the free edge of the cerebral falx
e) Drains into the internal jugular vein

43. Occipital sinus
a) It receives blood from the middle superficial cerebral vein
b) Lies in the attached border of the cerebellar falx
c) Drains into the transverse sinus
d) Runs in the free edge of the cerebral falx
e) Drains into the internal jugular vein

44. Superior petrosal sinus
a) It receives blood from the middle superficial cerebral vein
b) Lies in the attached border of the cerebellar falx
c) Drains into the transverse sinus
d) Runs in the free edge of the cerebral falx
e) Drains into the internal jugular vein

45. Inferior petrosal sinus
a) It receives blood from the middle superficial cerebral vein
b) Lies in the attached border of the cerebellar falx
c) Drains into the transverse sinus
d) Runs in the free edge of the cerebral falx
e) Drains into the internal jugular vein

46. The pallidum projects to the cerebral cortex through [appeared on 2002/02]
a) The VPL of thalamus
b) The VPM of the thalamus
c) The reticular nucleus of the thalamus
d) The ventral anterior nucleus of the thalamus
e) The DL of the thalamus

47. The corticomedial part of the amygdala receives input from the
a) Thalamus
b) Hypothalamus
c) Reticular formation

13. It controls conjugate eye movement reflexly in repsonse to head movement and to the position of the head in space [appeared on 2002/02]
a) Medial longitudinal fasciculus (MLF)
b) Gower's tract
c) Flesching's tract
d) Central tegmental tract
e) Spinothalamic tract

14. Which of the following tracts has strong fascilitating effects on motoneurons innervating antigravity muscles
a) Medial vestibularspinal
b) Ventral corticospinal
c) Rubrospinal
d) Tectospinal
e) Gracile

15. The pericallosal artery is a direct branch of the
a) Anterior cerebral
b) Middle cerebral
c) Posterior cerebral
d) Anterior communicating
e) Internal carotid

16. The inferior pertrosal sinuses are interconnected by the
a) Cavernous sinus
b) Superior sagittal sinus
c) Inferior sagittal sinus
d) Torcular
e) Basilar plexus

17. It contains the anterior cerebral and anterior communicating arteries [appeared on 2002/02]
a) Carotid cistern
b) Prepontine cistern
c) Quadrigeminal cistern
d) Lamina terminalis cistern
e) Supramarginal gyrus

18. The ventral spinocerebellar tract enters the cerebellum through the
a) Brachium conjunctivum
b) Tectal plate
c) Middle cerebellar peduncle
d) Restiform body
e) Juxtarestiform body

19. It curves around the upturned end of the sylvian fissure
a) Angular gyrus
b) Cingulate gyrus
c) Fusiform gyrus
d) Pars opercularis
e) Supramarginal gyrus

20. Rexed's lamina X is located in the
a) Posterior horn
b) Intermediate zone
c) Anterior horn
d) Comissural area

21. It is the initial center of the Papez circuit
a) Hippocampal formation
b) Cingulate gyrus
c) Cingulum
d) Amygdala
e) Thalamus

22. Damage to the parasympathetic fibers of the oculomotor would result in
a) Lateral strabismus
b) Miosis
c) Ptosis
d) Miosis and ptosis
e) Ptosis and pupil dialation

23. Internuclear ophthalmoplegia is part of
a) Locked-in syndrome
b) Weber's syndrome
c) Foville's syndrome
d) Millard-Gubler syndrome
e) Parinaud's syndrome

48. Guillain-Barre syndrome typically affects the [appeared on 2002/02]
a) Posterior horns
b) Lateral horns
c) Anterior horns
d) Internal capsule
e) Corona radiata

49. Neurons of the nucleus dorsalis of Clarke give origin to the
a) Spinothalamic tract
b) Gracile fasciculus
c) Dorsal spinocerebellar tract
d) Ventral spinocerebellar tract
e) Rostral spinocerebellar tract

50. Fibers from the corticospinal tract synapse with the
a) Inferior colliculus
b) Superior olive
c) Olive
d) Nucleus ambiguous
e) Superior colliculus

51. The fibers of the spinal nucleus and the trigeminal nerve synapse primarily with
a) VPL
b) VPM
c) Anterior nucleus
d) Reticular nucleus
e) Centromedian nucleus

52. The nucleusambiguous is related to the following functional components of the vagal system
a) SVE
b) SVE and GVE
c) SVA and GSA
d) GSA and GSE
e) SVA and GSE

53. A lesion of the right pyramid is going to result in [appeared on 2002/02]
a) Left exteroceptive deficit
b) Left proprioceptive deficit
c) Left motor deficit
d) Right motor deficit
e) Right exteroceptive deficit

54. The transverse pontine fibers enters the cerebellum through the [appeared on 2002/02]
a) Superior cerebellar peduncle
b) Pontine reticular formation
c) Medial reticulospinal tract
d) Middle cerebellar peduncle
e) Inferior cerebellar peduncle

55. A lesion of Broadman's area 39 is going to result in [appeared on 2002/02]
a) Ipsilateral neglect syndrome
b) Contralateral neglect syndrome
c) Ipsilateral motor deficit
d) Ipsilateral motor deficit
e) None of the above

56. The line of Gennari is located in the
a) Olfactory cortex
b) Somatosensory cortex
c) Visual cortex
d) Hearing cortex
e) Gustatory cortex

57. The main efferent connections of the hypothalamus are carried by the
a) Fornix
b) Thalamus
c) Subthalamus
d) Anterior commissure
e) Posterior commissure

58. It is essential for the formation of new memories and learning
a) Amygdala
b) Hippocampus
c) Subiculum
d) Cingulum
e) Dentate gyrus

59. Taste fibers of the IX cranial nerve have ther cell bodies in the
a) Nodose ganglion
b) Petrosal ganglion
c) Geniculate ganglion

60. Based on their functions, which of the following cranial nerve nuclei would you expect to find lateral to the sulcus limitans
a) Hypoglossal nucleus
b) Spinal trigeminal nucleus
c) Abducens nucleus
d) Nucleus of Edinger-Westphal

61. The large bundle of axons that runs upward from the cerebellum into the midbrain tegmentum is the [may have appeared on 2002/02]
a) Superior cerebellar peduncle
b) Middle cerebellar peduncle
c) Inferior cerebellar peduncle
d) Medial longitudinal fasciculus

62. Sensory nerve fibers carrying sensation from all the teeth enter the CNS at the
a) Spinal cord
b) Medulla
c) Pons
d) Midbrain
e) Cerebellum

63. This nerve has fibers that extend into three different sensory cranial nerve nuclei in the pons
a) Trigeminal
b) Abducens
c) Facial
d) Vestibulocochlear
e) Glossopharyngeal

64. Which of these cranial nerves DOES NOT provide any sensory or motor innervation to the tongue
a) VII
b) VIII
c) IX
d) X
e) XII

65. A stroke that destroys the lateral medulla (PICA syndrome) is most likely to cause
a) Altered eye movements
b) Paralysis of ipsilateral tongue
c) Facial paralysis
d) Dizziness and hearing problems on the side of the damage

66. Axons in the spinal cord white matter may have their cell bodies in any of the following locations EXCEPT
a) Dorsal horn
b) Paravertebral chain ganglia
c) Cerebral cortex
d) Intermediate gray matter

67. A patient has a meningioma growing in the vertebral canal at the level of the L4 vertebral body. Which spinal cord reflex might be reduced in intensity because of direct compression damage to motor neurons in the adjacent spinal cord segments
a) Flexor reflex-hand
b) Biceps reflex
c) Knee jerk reflex
d) Ankle jerk reflex
e) None of the above affected

68. Which one of the following spinal cord segments has the longest dorsal/ventral rootlets [email gmb1994@aol.com with the answer]
a) C2
b) T8
c) L5
d) S4

69. Brodmann's area 43 is related to the
a) Olfactory system
b) Gustatory system
c) Vestibular system
d) Proprioceptive system
e) Vagal system

70. It is the origin of the interstitiospinal tract
a) Nucleus of Perlia
b) Nucleus of Darkschewitz
c) Nucleus of cajal
d) Medial vestibular nucleus
e) Lateral vestibular nucleus

72. The ganglion of scarpa has the cell bodies of the
a) Facial nerve
b) Glossopharyngeal nerve
c) Vagus nerve
d) Hypoglossal nerve
e) Vestibular nerve

73. It forms the dorsal part of the wall of the 3rd ventricle [know all the borders of the 3rd ventricle, tentorium, tegmentum, tectum]
a) Thalamus
b) Epithalamus
c) Lamina terminalis
d) Tuber cinereum
e) Hypothalamus

74. Which of the following is not a component of the lumbar cistern [know cister]
a) CSF
b) Cauda equina
c) Spinal cord
d) Filum terminale
e) Subarachnoid space

75. The facial colliculus is formed by fibers of the facial nerve surrounding the nucleus of the
a) Oculomotor nerve
b) Trochlear nerve
c) Vestibular nerve
d) Glossopharyngeal nerve
e) Abducens nerve

76. Medial striate arteries [appeared on 2002/02]
a) Anterior cerebral artery
b) Middle cerebral artery
c) Posterior cerebral artery
d) Basilar artery
e) Internal carotid artery

77. Lateral striate arteries [appeared on 2002/02]
a) Anterior cerebral artery
b) Middle cerebral artery
c) Posterior cerebral artery
d) Basilar artery
e) Internal carotid artery

78. Thalamoperforating arteries [appeared on 2002/02]
a) Anterior cerebral artery
b) Middle cerebral artery
c) Posterior cerebral artery
d) Basilar artery
e) Internal carotid artery

79. Thalamogeniculate arteries [appeared on 2002/02]
a) Anterior cerebral artery
b) Middle cerebral artery
c) Posterior cerebral artery
d) Basilar artery
e) Internal carotid artery

80. Circumflex arteries [appeared on 2002/02]
a) Anterior cerebral artery
b) Middle cerebral artery
c) Posterior cerebral artery
d) Basilar artery
e) Internal carotid artery

81. It runs from the posterior end of the cavernous sinus to the transverse sinus
a) Superior petrosal sinus
b) Straight sinus
c) Inferior petrosal sinus
d) Sphenoparietal sinus
e) Temporal sinus

83. A lesion affecting the right medial lemniscus results in
a) Right-sided loss of tactile discrimination
b) Left-sided loss of pain and temperature
c) Right-sided loss of pain and temperature
d) Left-sided loss of tactile discrimination

84. The surgical treatment of the tethered cord syndrome is transection of the
a) Denticulate ligaments
b) Cauda equina
c) Conus medullaris
d) Filum terminale
e) Coccygeal ligament

85. Spinal shock is a classical finding in patients with
a) Anterior cord syndrome
b) Central cord syndrome
c) Dorsolateral cord syndrome
d) Transection of spinal cord
e) Brown-sequard syndrome

86. It carnes the SVA fibers of the facial nerve [know Greater petrosal nerve, Chorda tympani nerve]
a) Greater petrosal nerve
b) Chorda tympani nerve
c) Lingual nerve
d) Zygomatic nerve

87. Cell bodies of fibers innervating the cornea are located in the
a) Gasserian ganglion
b) Mesencephalic nucleus
c) Main sensory nucleus
d) Nucleus of the spinal tract
e) Maxillary nerve

88. The optic tract passes through the
a) Anterior limb of the internal capsule
b) Posterior limb of the internal capsule
c) Sublenticular part of the internal capsule
d) Retrolenticular part of the internal capsule
e) Genu of the internal capsule

89. The ciliospinal center is located at
a) C8 and T1 spinal cord segments
b) C5 and C7 spinal cord segments
c) T8 and T10 spinal cord segments
d) L1 and L2 spinal cord segments

90. Which of the following structures forms the acoustic tubercle
a) Lateral cochlear nucleus
b) Ventral cochlear nucleus
c) Restiform body
d) Dorsal cochlear nucleus
e) Juxtarestiform body

91. As it decussates it forms the trapezoid body [appeared on 2002/02]
a) Lateral lemniscus
b) Central tegmental tract
c) Ventral acoustic stria
d) Dorsal acoustic stria
e) Medial lemniscus

92. The nucleus dosalis of Clarke is located in
a) The anterior horn of T1 - L2
b) The base of the posterior horn of T1 - L2
c) The base of the posterior horn of T1 - L2
d) The lateral horn
e) The ventral white matter

96. The transverse temporal gyri of Herschl contain cell bodies of neurons whose primary function is
a) Visual
b) Auditory
c) Sensory somesthetic
d) Visceral sensory
e) Taste

97. Transection of lower motor neurons LEAST likely result in
a) Babinski's sign
b) Fibrillations
c) Areflexia
d) Muscle atrophy
e) Flaccid paralysis

98. The following is the site of emergence of the IX cranial nerve
a) Pontomedullary suclus
b) Cerebellopontine angle
c) Floor of the IV ventricle
d) Preolivary suclus
e) Postolivary sulcus

99. Which of the following gyri is located between the inferior temporal sulcus and the collateral fissure [appeared on 2002/02; Fusiform gyrus and Hippocampal gyrus answer choices were inverted]
a) Fusiform gyrus
b) Hippocampal gyrus

100. The following structure separates the lentiform nucleus from the head of the caudate nucleus [not sure if this appeared on 2002/02]
a) Thalamus
b) Lateral ventricle
c) Posterior limb of the internal capsule
d) External capsule
e) Anterior limb of the internal capsule

101. The straight sinus drains into the
a) Confluence of the sinuses
b) Superior petrosal sinus
c) Inferior petrosal sinus
d) Transverse sinus
e) Great vein of Galen

102. It contains the vein of Galen [know the borders of the Quadrigeminal cistern]
a) Crural cistern
b) Ambiens cistern
c) Supracerebellar cistern
d) Sylvian cistern
e) Quadrigeminal cistern

103. Brodman's area 22 [know Brodman's area 39]
a) Apraxia
b) Ageusia
c) Fluent aphasia
d) Tactile agnosia
e) Loss of initiative and judgement

104. Brodman's area 43 [know Brodman's area 39]
a) Apraxia
b) Ageusia
c) Fluent aphasia
d) Tactile agnosia
e) Loss of initiative and judgement

105. Brodman's area 6 [know Brodman's area 39]
a) Apraxia
b) Ageusia
c) Fluent aphasia
d) Tactile agnosia
e) Loss of initiative and judgement

106. Prefrontal cortes [know Brodman's area 39]
a) Apraxia
b) Ageusia
c) Fluent aphasia
d) Tactile agnosia
e) Loss of initiative and judgement

107. Superior parietal lobule [know Brodman's area 39]
a) Apraxia
b) Ageusia
c) Fluent aphasia
d) Tactile agnosia
e) Loss of initiative and judgement

108. The following tract has autonomic parasympathetic projections
a) Corticorubral
b) Rubrospinal
c) Tectospinal
d) Medial reticulospinal
e) Lateral reticulospinal

109. Most of the symptoms and signs seen in patients with upper motoneuron lesions are associated with increased reactivity of
a) The internal capsule
b) Pyramidal cells
c) Alpha motoneurons
d) Beta motoneurons
e) Gamma motoneurons

110. The following has connections with the striatum via the external capsule
a) Cerebral cortex
b) Thalamus
c) Substantia nigra
d) Subthalamic nucleus
e) Caudate nucleus

111. An aneurysm arising from the posterior communicating segment of the internal carotid artery will produce
a) Ipsilateral III cranial nerve palsy
b) Contralateral III cranial palsy
c) Ipsilateral VI cranial nerve palsy
d) Contralateral VI cranial nerve palsy
e) Ipsilateral trochlear nerve palsy

112. CSF is absorbed at the
a) Superior sagittal sinus
b) Lateral ventricle
c) Third ventricle
d) Interior sagittal sinus
e) Arachnoid granulations

113. A lesions affecting the conus medullaris will produce
a) Nonal bladder fuction
b) Spastic bladder
c) Flaccid bladder
d) Urinary incontinence

114. Has connections with limbic system
a) Corticospinal tract
b) Corticobulbar tract
c) Spinothalamic tract
d) Spinoreticular tract

115. The spinal accessory nucleus is located in the [know nucleus]
a) Caudal part of the medulla
b) Rostral part of the medulla
c) Anterior horn C2 to C5
d) Anterior horn C5 to C7
e) Pontine tegmentum

116. The cells in the inferior salivatory nucleus send their axons via the
a) Intermediate nerve
b) Vagus nerve
c) Glossopharyngeal nerve
d) Pterygopalatine ganglion
e) Spinal nerve

117. Which of the following arteries is the most common site for berry aneursym
a) Anterior communicating
b) Posterior communicating
c) Anterior choroidal
d) Anterior cerebral
e) Posterior cerebral

119. The main contribution from midbrain structures to the central tegmental fasciculus comes from [there is a similar question on 2002/02]
a) Substantia nigra
b) Edinger-Westphal nucleus
c) Nucleus of Darkschewitz
d) Red nucleus
e) Locus cereleus

120. Which tract is the main mediator of nystagmus and conjugated gaze
a) Central tegmental fasciculus
b) Lateral vestibulospinal tract
c) Dorsal longitudinal fasciculus
d) Tectospinal tract
e) Medial longitudinal fasciculus

121. Binasal hemianopia results from
a) Calcified internal carotid artery
b) Midsagittal transection of the optic chiasm
c) Bilateral transection of Meyer's loop
d) Bilateral occlusion of the calcarine artery
e) None of the above

122. Which of the following nuclei is the main origin of the ansa lenticularis [very similar question to this appeared on 2002/02]
a) Substantia nigra
b) Kolliker-fuse
c) Luys'
d) Globus pallidus
e) Nageotte's

123. Phineas Gage syndrome results from a lesion where
a) Insula
b) Areas 39 & 40
c) Areas 28 & 34
d) Prefrontal lobe
e) Nucleus accumbens

124. Romberg's sign is seen in lesions of
a) Lateral reticulospinal tract
b) Dorsal column
c) Ventral corticospinal tract
d) Lateral reticulospinal tract
e) Rubrospinal tract

125. The terminal branch of the middle cerebral artery supplies the
a) Superior temporal gyrus
b) Angular gyrus
c) Area 40
d) Postcentral gyrus
e) Primary motor area

126. Ipsilateral deafness results by destruction of
a) Dorsal and ventral cochlear nuclei
b) Corti's ganglion
c) Cochlear nerve
d) Cochlea
e) All of the above

127. Kolliker-Fuse nucleus is best seen in the
a) Midbrain
b) Pons
c) Medulla
d) Subthalamus
e) Zona incerta

128. It is the shortest part of the brainstem and contains centers associated with the auditory and visual systems
a) Diencephalon
b) Pons
c) Medulla
d) Midbrain
e) Basal ganglia

129. The pineal gland and the habenula are located in the [appeared on 2002/02]
a) Thalamus
b) Hypothalamus
c) Epithalamus
d) Subthalamus
e) Metathalamus

131. The pyriform cortex is related to
a) Auditory system
b) Visual system
c) Olfactory system
d) Gustatory system
e) Tactile system

132. Which of the following cranial nerves is related to the diencephalon [appeared on 2002/02]
a) I
b) II
c) VIII
d) V
e) VII

133. From the VPL nucleus of the thalamus, fibers of the spino thalamic tract reach the
a) Precentral gyrus
b) Superior frontal gyrus
c) Middle frontal gyrus
d) Postcentral gyrus
e) Superior temporal gyrus

134. Corticostriate projections reach the caudate and putamen via the [very similar to a question on 2002/02]
a) Posterior limb of internal capsule
b) Globus pallidus
c) Thalamus
d) Substantia nigra
e) External capsule

135. The cell bodies of the dorsal root fibers are located in the
a) Posterior horn
b) Intermediate zone
c) Dorsal root ganglia
d) Dorsal root entry zone
e) Anterior zone

136. The rostral part of the solitary nucleus is related to [know solitary nucleus in relation to nerves]
a) Taste
b) Secretion of tears
c) Secretion of saliva

137. Prinaud's syndrome
a) Tectal plate
b) Dorsolateral sector of the medulla
c) Anterior aspect of the midbrain
d) Internuclear ophthalmoplegia
e) Facial and abducens paralysis

138. Weber's syndrome
a) Tectal plate
b) Dorsolateral sector of the medulla
c) Anterior aspect of the midbrain
d) Internuclear ophthalmoplegia
e) Facial and abducens paralysis

139. Foville's syndrome
a) Tectal plate
b) Dorsolateral sector of the medulla
c) Anterior aspect of the midbrain
d) Internuclear ophthalmoplegia
e) Facial and abducens paralysis

140. Wallenberg's syndrome
a) Tectal plate
b) Dorsolateral sector of the medulla
c) Anterior aspect of the midbrain
d) Internuclear ophthalmoplegia
e) Facial and abducens paralysis

142. A lesion of the supraoptic and paraventricular of the hypothalamus nuclei results in [know diabetes insipidus]
a) Diabetes insipidus
b) Growth retardation
c) Hypogonadism
d) Visual loss
e) Diabetes mellitus

143. The major output of the cerebellum goes through the [know Inferior cerebellar peduncle, Middle cerebellar peduncle, Superior cerebellar peduncle]
a) Inferior cerebellar peduncle
b) Middle cerebellar peduncle
c) Superior cerebellar peduncle
d) Interposed nuclei
e) Fastigial nucleus

144. Asterognosis is a prominent sign in lesion of the
a) Anterolateral system
b) Corticospinal tract
c) Spinoreticular tract
d) Cuneate tract
e) Lateral column of the spinal cord

145. A lucunar stroke affecting the right subthalamic nucleus produces
a) Right chorea
b) Left chorea
c) Right hemiballismus
d) Left hemiballismus
e) Right athetosis

146. The right and left amygdala are connected by the [appeared on 2002/02]
a) Anterior commissure
b) Posterior commissure
c) Corpus callosum
d) Forceps major
e) Forceps minor

147. It is a branch of the internal carotid
a) Posterior cerebral artery
b) Posterior choroidal artery
c) Anterior choroidal artery
d) Pericallosal artery
e) Precentral artery

148. The cardinal symptom in syringomyelia is [appeared on 2002/02]
a) Loss of proprioception
b) Bilateral motor deficit
c) Urinary incontinence
d) Loss of stretch reflexes
e) Sensory dissociation

149. The components of the MLF are [appeared on 2002/02]
a) Medial reticulospinal, medial, vestibulospinal, interstitiospinal and tectospinal tracts

150. The cortical exteroceptive representation of the perineum is located in the
a) Precentral gyrus
b) Superior parietal lobule
c) Postcentral gyrus
d) Paracentral lobule
e) Interior parietal lobule

151. The SVE functional component of the trigeminal nerve is carried by the [know SVE]
a) Ophthalmic division
b) Maxillary division
c) Mandibular division
d) Only "Ophthalmic division" and "Mandibular division"
e) Only "Maxillary division" and "Mandibular division"

152. Which of the following interrelates the midbrain with the cerebellum
a) Brachium pontis
b) Restiform body
c) Brachium conjunctivum
d) Superior colliculus
e) Inferior colliculus

154. The most conspicuous structure in the middle part of the pons is the
a) Transverse pontine fibers
b) Floor of the IV ventricle
c) Trigeminal nucleus
d) Red nucleus
e) Trochlear nucleus

155. The median eminence is located in the [know medullary part, pontine part]
a) Floor of the IV ventricle (medullary part)
b) Tegmentum of the pons
c) Tectal plate
d) Floor of the IV ventricle (pontine part)
e) Medulla

156. Which of the following cranial nerves has the longest intracranial course
a) Facial
b) Oculomotor
c) Abducens
d) Trochlear
e) Vagus

157. It has the least amount of gray matter
a) Cervical segment
b) Thoracic segment
c) Lumbar segment
d) Sacral segment

158. The forceps minor is formed by fibers of the [know forceps minor]
a) Splenium of the corpus callosum
b) Body of the corpus callosum
c) Amygdala
d) Rostrum of the corpus callosum
e) Interthalamic adhesion

159. Clonus is seen in patients with
a) Lower motoneurons lesion
b) Basal ganglia lesions
c) Hypthalamic damage
d) Upper motoneuron lesions
e) Guillan-Barre syndrome

160. The most commonly affected in lateral transtentorial herniations
a) Internal carotid artery
b) Anterior cerebral artery
c) Middle cerebral artery
d) Posterior cerebral artery
e) Basilar artery

161. It is located beneath the uncus near the dorsomedial tip of the temporal lobe
a) Septal region
b) Fusiform gyrus
c) Cingulate gyrus
d) Fornix
e) Amygdala

162. The hippocampal formation receives input from all parts of the cortex through
a) Thalamus
b) Hypothalamus
c) Amygdala
d) Septal region
e) Cingulum

163. Lesion of Brodmann's area 44 and 45 results in
a) Ipsilateral motor deficit
b) Contralateral motor deficit
c) Fluent aphasia
d) Nonfluent aphasia

164. It lies adjacent to the tentorial edge and contains the IV and VI cranial nerves [may have appeared on 2002/02]
a) Crural cistern
b) Ambient cistern
c) Quadrigeminal cistern
d) Supracerebellar cistern
e) Premedullary cistern

166. It runs from the posterior end of the cavernous sinus to the transverse sinus
a) Superior petrosal sinus
b) Straight sinus
c) Inferior petrosal sinus
d) Sphenoparietal sinus
e) Temporal sinus

167. Which of the following tracts has strong facilitating effects on motoneurons innervating antigravity muscles
a) Medial vestibulospinal
b) Ventral corticospinal
c) Rubrospinal
d) Tectospinal
e) Gracile

168. Most fibers of the optic tract end in the
a) Medial geniculate bodies
b) Lateral geniculate bodies
c) Loop of Meyer
d) Visual cortex
e) Occipital lobe

169. Conscious proprioception of the leg is carried by the
a) Fasciculus cuneatus
b) Fasciculus gracilis
c) Spinothalamic tract
d) Dorsal spinocerebellar tract

170. Rexed lamina VII is located in the
a) Posterior horn
b) Posterior horn and intermediate zone
c) Intermediate zone only
d) Anterior horn and interior zone
e) Comissural area

171. The inability to recognize the posterior of the limbs is a prominent sign of lesion affecting the [email gmb1994@aol.com with the best answer]
a) Spinothalamic tract
b) Corticospinal tract
c) Trigeminovascular system
d) Fasciculus gracilis

172. Spasticity is a prominent sign in lesions affecting the
a) Posterior horns of spinal cord
b) Tectospinal tract
c) Spinothalamic tract
d) Cuneocerebellar tract
e) Corticospinal tract

173. Additional remembered topics related to this quizbank: [these choices do not have answers ... email gmb1994@aol.com with yoru suggested answers]
a) What are the lesions affecting exteroception in limbs
b) Know the origens of the cranial nerves
c) Which nerve comes out at the Cerebellar Pontine angle
d) Know the tracts
e) Know what nerves descussate through the thalamus and where they originate and terminate
f) Know reticular formation functions and borders
g) Know striatum and where they orginate and terminate
h) What are the borders of the Quadrigeminal cistern
i) Know Brodmann's area 39
j) What is the innervation to the superior oblique
k) Know olives and their borders
l) Know vagal nerve (general and special designations ... SVE, GVE, GVA)
m) Know figure 19 on page 29 and figure on page 28 from Ridiculously Simple Neuroanatomy
n) Know about the thalamus and all lemniscii related (Netter plate 104-105)
o) Know lesions to Edin-Westphal and left optic nerve which results in left eye ptosis (a, b or both)
p) Many nucleus locations
q) What is the tectum made of
r) I did not see any cell body questions on 2002/02
s) Know all borders of 3rd ventricle
t) Know the borders of the tentorium, tegmentum, tectum
u) Know the contents of the facial canal ... 2 questions related to chorda tympani ... questions related to all petrosals
v) Know the contents of the cisterns
w) Know all the nerves associated with the taste
x) Know the innervation of the digestive muscles (anterior and posterior)
y) How is the amygdala attached

1. (b)
2. (c)
3. (e)
4. (c)
5. (c)
6. (e)
7. (d)
8. (c)
9. (e)
10. (e)
11. (b)
12. (d)
13. (a)
14. (a)
15. (a)
16. (e)
17. (d)
18. (a)
19. (e)
20. (d)
21. (a)
22. (e)
23. (c)
24. (d)
25. (d)
26. (a)
27. (a)
28. (c)
29. (e)
30. (a)
31. (c)
32. (d)
33. (c)
34. (d)
35. (a)
36. (c)
37. (a)
38. (d)
39. (d)
40. (c)
41. (d)
42. (a)
43. (b)
44. (c)
45. (e)
46. (d)
47. (d)
48. (c)
49. (c)
50. (e)
51. (b)
52. (a)
53. (c)
54. (d)
55. (b)
56. (c)
57. (a)
58. (b)
59. (e)
60. (a)
61. (a)
62. (c)
63. (c)
64. (b)
65. (a)
66. (d)
67. (e)
68. (c)
69. (b)
70. (c)
71. (b)
72. (e)
73. (a)
74. (c)
75. (e)
76. (a)
77. (b)
78. (c)
79. (c)
80. (d)
81. (a) (b)
82. (a)

22. In the TCA cycle:

a) Four NADH molecules are produced
b) Substrate-level phosphorylation occur by the action of sccinyl CoA thiokinase
c) The reactions can proceed in the absence of molecular oxygen
d) Fumarase reactions occur immediately after isocitrate dehydrogenases reactions
e) Glutamate dehydrogenase is a key enzyme

23. In fructose metabolism:

a) Deficiency of hexokinase results in fructosuria
b) Glucokinase is the major enzyme
c) Fructokinase phosphorylates fructose
d) Fructose 6 phosphate is produced by fructokinase
e) Aldolase A is the key enzyme

24. Oxidation of glucose by red blood cells gives:

a) Lactate + NADH+H
b) 36 ATP molecules per molecule of glucose
c) Lactate and NAD+
d) Carbon dioxide and water
e) 2 NADPH

25. The pentose phosphate pathway:

a) Produces ATP
b) Is a mitochondrial pathway
c) Is important for reductive biosynthesis
d) Is least active in the liver
e) Is inhibited by high glucose levels

26. The following agent interferes with oxidative phosphorylation:

a) Carbon monoxide inhibits cytochrome reductase
b) Cyanide inhibits NADH-dehydrogenase
c) Rotenone enhances electron transfer
d) Oligomyein inhibits electron transfer and phosphorylation
e) Dintrophenol enhances ATP synthesis

27. UDP-Glucose:

a) Is indirectly involved in bilirubin conjugation
b) Is a substrate for the enzyme 4-epimerase in lactating mammary gland
c) Is intermediate in sphingolipid synthase
d) Is required for galactose metabolism
e) All of the above

28. Glycogen synthesis:

a) Is a mitochondrial pathway
b) Is activated during the fasting state
c) Occurs in the liver and muscle
d) Produces uridine diphosphate from glycogen
e) All of the above

29. Pyruvate is:

a) Converted to acetyl CoA by pyruvate carboxylase
b) Four-carbon compound
c) Converted to oxaloacetate by pyruvate dehydrogenase complex
d) Converted to alanine by transamination
e) Intermediate in the TCA cycle

30. In glycogen storage diseases:

a) Fatigability and muscle cramps are main features of Von Geirke's disease
b) Hypoglycemia is the main feature of type V
c) Type II is associated with glucose 6 phosphatase deficiency
d) Type IV is called Andersen's disease
e) Hepatomegally is the main feature of all types

31. The following enzyme would be impaired in biotin deficiency:

a) Fructose 1,6-bisphosphate
b) Pyruvate kinase
c) PEP carboxykinase
d) Pyruvate carboxylase
e) Malate dehydrogenase

32. The following statement about the pentose phosphate pathway is correct:

a) Pentoses can be formed both oxidatively and non-oxidant
b) This pathway is important to fatty acid synthesis because it produces equivalents
c) Fructose 6 phosphate can be used to make ribose-5-phosphate
d) Glyceraldehyde-3-phosphate can be used to make dinitrophenol
e) All of the above are correct

33. The following is caused by deficiency in aldolase B:

a) Hereditory fructose intolerence
b) Jaundice
c) Hypoglycemia
d) Catarct
e) Ataxia

34. The rate of glycolysis in the liver is increased by:

a) Insulin
b) ATP
c) Citrate
d) NADH
e) Glucose 6-phosphate

35. The preparative step of glycolysis involves the following:

a) ATP synthesis at the substrate level
b) The incorporation of P_i into a triose phosphate
c) The reduction of NAD+ to NADH
d) The formation of 1,3-bisphosphoglycerate
e) Formation of triose phosphates from a hexose diphosphorase

36. The following is present in the liver, but is absent in muscles:

a) Pyruvate carboxylase
b) Glycogen synthase
c) Lactate dehydrogenase
d) Pyruvate dehydrogenase
e) Glucose-6-phosphatase

37. The following enzyme utilizes FAD as a coenzyme:

a) Malate dehydrogenase
b) Aconitase
c) Succinate dehydrogenase
d) Isocitrate dehydrogenase
e) Lactate dehydrogenase

38. A patient complaining of painful cramps in the legs during exercise and has easy fatigability, the most likely cause is:

a) Diabetes mellitus
b) Deficiency of glucose-5-phosphatase
c) Deficiency of the debranching enzyme or muscle phosphorylase
d) Defective glycogen synthesis
e) Over-storage of glycogen

39. Malonate inhibits succinate dehydrogenase because:

a) Binds irreversibly to the active site
b) Covalently modifies the enzyme
c) Resembles succinate but cannot react
d) Displaces he FAD coenzyme
e) Chelates a metal ion required by the enzyme

40. In oxidative phosphorylation:

a) An electrochemical potential is created across the inner mitochondrial membrane
b) ATP is synthesized from ADP and Pi
c) Protons are pumped into the interspace between the inner and outer mitochondrial membranes
d) The energy released from the electron flow is directly used for synthesis of ATP
e) D is incorrect

41. Phosphofructokinase-1:

a) Is activated by high level of ATP
b) Catalyzes the only irreversible reaction in glycolysis
c) Is activated by the high level of AMP
d) Converts fructose 1-phosphate to fructose 1,6-bisphosphate
e) Is costimulated with fructose 1,6 bisphosphatase

42. In gluconeogenesis:

a) Pyruvate is converted to acetyl CoA
b) Valine and Leucine are substrates
c) Energy consumption is high
d) Insulin is needed to enhance the synthesis of glucose
e) Acetyl CoA is a substrate